DECO7230 · DESIGN CONCEPT REPORT · SUBMISSION 1

XR Renovation Previewer

Redesigning the phone paint visualiser as an embodied XR tool

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Design activity documented at **github.com/kaikenehme/deco7230-xr**

01 THE APPLICATION, AND WHY IT NEEDS XR

CATEGORY: CREATION

I am redesigning the **phone renovation preview app**: Dulux Visualizer, Houzz, IKEA Kreativ, Home Depot Project Color. These are interior design and space planning tools, where the user builds a scheme for a room they are about to change. That puts them in the Creation category.

They all fail the same way. A candidate colour or finish is shown as a flat rectangle on a six inch screen, lit by a generic render, with the real room nowhere in sight. The three things that decide whether a renovation works are all spatial:

Scale	A colour looks different across a whole wall than it does in a thumbnail. The phone gets this wrong every time.
Adjacency	What matters is how a candidate sits *next to* the floor, sofa or benchtop you are keeping. A swatch grid never shows that.
Light	The thing that ruins real renovations. The phone shows one fixed lighting condition. A room has several, and they change through the day.

All three only exist when you are standing in the space at full size, which is why this is a redesign and not a port. XR can supply what the phone cannot.

02 USER TASKS AND GOALS

	TASK	GOAL
T1	Distinguish what is staying in the room from what is changing	Set up the constraints every scheme has to satisfy. Most renovations are not blank slates
T2	Evaluate a candidate finish against a fixed element, in place, at full size, under the room's own light	Judge one choice with the context that decides it, instead of judging it in isolation
T3	Assemble and compare complete schemes that all respect that same constraint	Commit to a whole room direction instead of a run of unrelated single colours

03 THE XR CONCEPT

Immersive environment. One room inside a house. In the early prototypes it is a modelled virtual living room. In the final prototype it is the user's *own* room seen through Meta Quest passthrough, so the fixed elements they design around are the real ones.

The central idea. You never choose from a catalogue. You **pull a sample straight off something you are keeping**, like the timber floor or the sofa, and that sample generates the range of finishes that work with it. The constraint already sitting in the room is what produces the options. That kills the menu and most of the choice paralysis at the same time.

STEP 01	STEP 02	STEP 03	STEP 04	STEP 05
Mark	**Pull**	**Hold up**	**Tune**	**Commit**
Sweep an open hand across each surface to tag it as staying or changing.	Touch a kept surface and draw your hand away. A sample of it comes with you.	Raise it toward a changeable surface. As it nears, that surface previews the colour.	Twist your wrist to move through the range that works with the source. Warmer, cooler, deeper.	Release. Move away without releasing and the preview reverts.

No menu appears anywhere in that sequence. Lighting works the same way: you touch a lamp or a wall switch to cycle warm bulb, cool bulb and daylight. Every control is an object in the room, not an interface floating over it.

04 XR INTERACTIONS AND AFFORDANCES

AFFORDANCE	WHAT THE USER DOES	WHY THE PHONE CANNOT DO THIS
Full scale judgement	Stands in the room and sees a candidate across an entire wall	Colour perception depends on scale. A thumbnail is not a small version of the same experience
Physical adjacency	Holds a sample so it visually touches the floor being kept	A phone can only put them side by side in a grid, never in contact in real space
Real and variable light	Touches a lamp to switch between warm bulb, cool bulb and daylight	The app renders one fixed lighting condition, so the decision gets made under light that will never exist
Proprioceptive comparison	Holds the sample at arm's length, moves it into shadow, into the corner	People already judge samples this way in shops. A touchscreen cannot use the arm at all
Embodied parameter control	Rotates the wrist to travel a constrained colour range	Swaps a scrollable palette for one continuous movement through options already known to work
Viewpoint by locomotion	Walks to where the sofa sits and looks back at the wall	Colour and finish shift with viewing angle and distance. A fixed camera hides all of it

05 IDEATION AND REFINEMENT

The first idea was just *"recolour the walls of a room in VR."* Two rounds of refinement got it to the concept above.

Round 1: low fidelity physical prototype

I built a living room out of paper, cardboard and playdoh in the Week 2 Studio: two hinged paper walls scribbled red and green, a bare kraft card armchair, a playdoh sofa, a card table and two LED tea lights. Building it turned up two things I had not planned for.

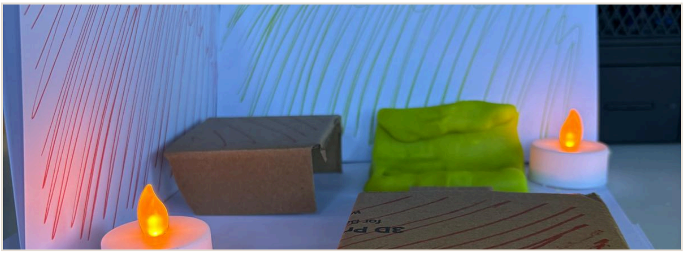

Plate 1. The same red paper reads warm orange beside the flame and cool pink away from it. One surface, two apparent colours.

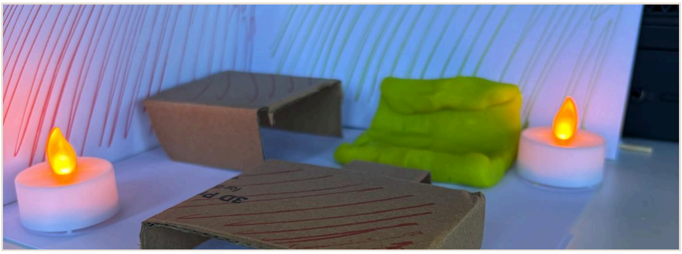

Plate 2. At eye level. The playdoh sofa was matched to the green wall and the card table to the red one, unprompted.

Light	The tea lights were an afterthought and ended up the most useful part of the build. They showed the concept's central claim, that apparent colour depends on light, in cardboard and before any code existed. I moved light control from "out of scope" to a core interaction.
Matching	I coordinated the furniture to the wall colours by hand while building, without deciding to. The behaviour the concept is meant to support showed up on its own. Weak evidence, but real.

Round 2: peer critique in Studio

Peers made three requests, all phrased as "add a button". Two had a better answer underneath.

Light on/off	**Adopted, reframed.** Built as a lamp you touch instead of a menu button, and extended to three states so that "does this green survive a warm bulb at 9pm?" becomes answerable.
Rotate furniture	**Adopted, deferred.** I first read this as an artefact of tabletop scale, since you cannot walk around a model on a desk. Checking with the people who asked showed they meant real layout change, so it becomes its own system in a later prototype instead of a control bolted onto this one.
Switch VR / MR	**Adopted, promoted.** Originally a one way progression across the three prototypes. Reframed as a live switch, so a scheme can be compared in the real room against an idealised virtual version of it.

Nobody challenged the core loop in either round, so it carries forward unchanged.

06 INITIAL TESTING PLAN

What will be tested. Two things. First, the sample loop (pull from a kept surface, hold up to preview, twist to tune, release to commit). Second, the constraint mechanism that generates the options out of those kept surfaces. Scheme comparison, layout editing and passthrough are out of scope for this first round.

	ASSUMPTION	WHAT WOULD INVALIDATE IT
A1	**Discoverability.** The loop can be performed without instruction	Participants tap or press surfaces, hunt for a menu, or need prompting before completing a single change
A2	**Constraint as help.** Generating options from what is staying reduces choice paralysis instead of frustrating people	Participants report the options as restrictive, ask for a full palette, or abandon the constraint when offered the chance

A2 is the one that matters. The whole concept rests on it and it can fail. If it does, that is a finding worth having, not a setback.

Data to be collected

MEASURE	TYPE	METHOD
Time to first unprompted successful change	quantitative	Observer, stopwatch
Number of facilitator prompts required	quantitative	Observer tally
Whether twist to tune is discovered unprompted	boolean	Observation
Samples tried before committing to one	quantitative	Observer count
Confidence in the final choice, 1 to 5	quantitative	Post task question
Whether the limited option set felt helpful or restrictive, and why	qualitative	Post task question. **Primary evidence for A2**
Reasons given for rejecting a candidate	qualitative	Think aloud, recorded verbatim

Method. Think aloud, with no explanation of the interface beforehand, because discoverability cannot be measured once you have described it. Five minutes per participant, five participants minimum, run in the Studio session with peers and a tutor, screen recorded from the headset with consent. I am leaving out standardised instruments like SUS and presence questionnaires at this stage. They are built for longer sessions and would report noise at five minutes.